MILITARY SPECIFICATION

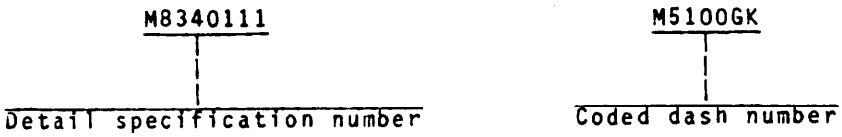
RESISTOR-CAPACITOR NETWORK, FIXED FILM RESISTORS AND CERAMIC CAPACITOR, STYLE RZ110, 16 PIN DIP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RZ110, 16 pin dual-in-line package (DIP) resistor-capacitor network.

1.2 Military part number. Networks covered by this specification shall be identified by a military part number which shall consist of the basic number of this specification and a coded dash number. The military part number shall be in the following form:



The coded dash number shall be derived in accordance with MIL-R-83401 and this specification.

1.2.1 Schematic. The schematic of the resistor-capacitor network shall be identified by a single letter in accordance with figure 1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-R-83401	-	Resistor Networks, Fixed, Film, and Resistor-Capacitor Networks, Fixed Film Resistors and Ceramic Capacitor, General Specification for.
MIL-R-83401/2	-	Resistor Network, Fixed, Film, Style RZ020, 16 Pin DIP.

(Copies of the specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Aeronautical Systems Division, (ASD/ENESS), Wright-Patterson AFB, OH 45433-6503 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-83401 and as specified herein.

3.2 Design and construction. The network shall be of the design, construction, and physical dimensions as specified on figure 2.

3.3 Resistor requirements.

3.3.1 Resistance. Resistance values are specified in table I.

3.3.2 Resistance tolerance. Unless otherwise specified, see table I, resistors are available in resistance tolerances of ±1 percent (F), ±2 percent (G), and ±5 percent (J).

3.3.3 Resistance characteristics. Networks are available in resistance characteristics K and M.

3.3.4 Power rating. The power rating shall be 0.1 watt for each resistor. Total network power rating shall be 1.2 watts.

3.3.5 Voltage rating. The maximum continuous working voltage for each resistor shall not exceed 100 volts dc or ac rms.

3.4 Capacitor requirements.

3.4.1 Capacitance. When tested as specified in 4.2.1, the capacitance value shall be 0.01 µF, within tolerance.

3.4.2 Capacitance tolerance. Capacitors are available in capacitance tolerance of ±20 percent (M).

3.4.3 Dissipation factor. When tested as specified in 4.2.2, the dissipation factor shall not exceed 2.5 percent.

3.4.4 Voltage ratings. The voltage rating shall be 25 volts dc.

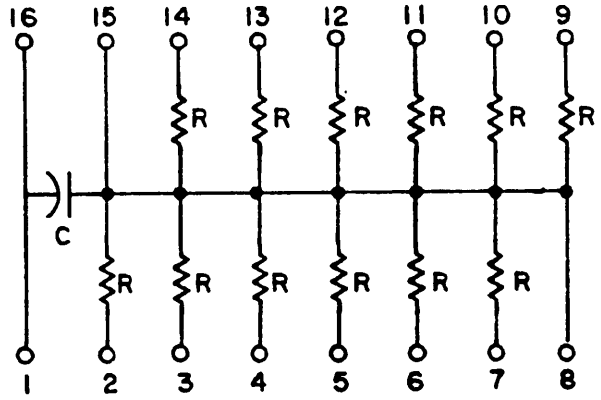
TABLE I. Resistance and capacitance values.

Part number 1/ M8340111	Resistance (ohms)		Capacitance (microfarads)
Schematic K			
-5100-K 1/	510		0.01
Schematic M			
-50R0-M 1/	50		0.01
-68R0-M	68		0.01
-75R0-M	75		0.01
-1000-M	100		0.01
Schematic N			
	R_S	R_E	
-A001XN 1/	39 ±5 percent	500 ±2 percent	0.01

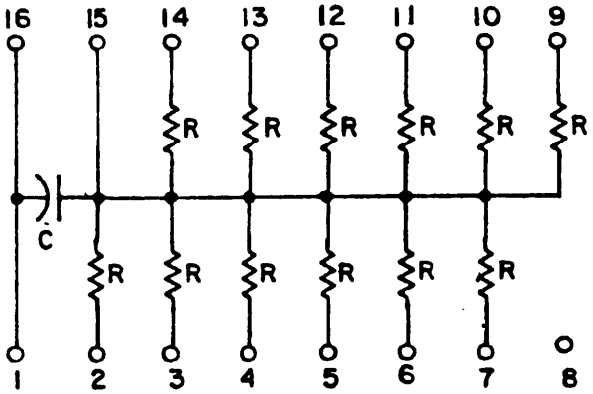
1/ Dashes in part number shall be coded in accordance with MIL-R-83401.

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SCHEMATIC "K"



SCHEMATIC "M"



SCHEMATIC "N"

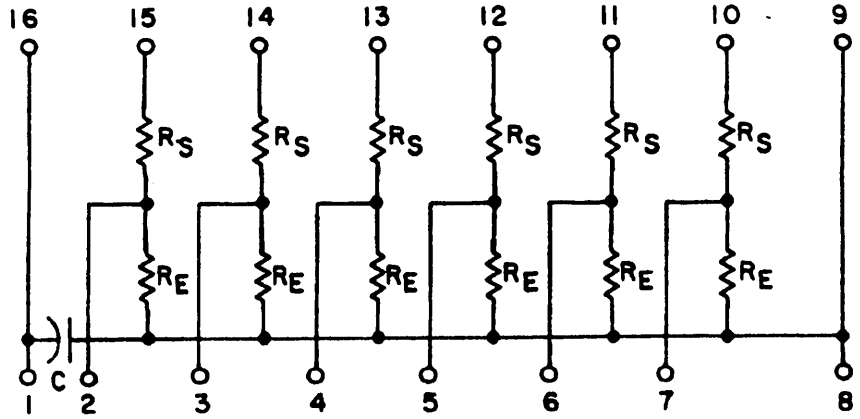
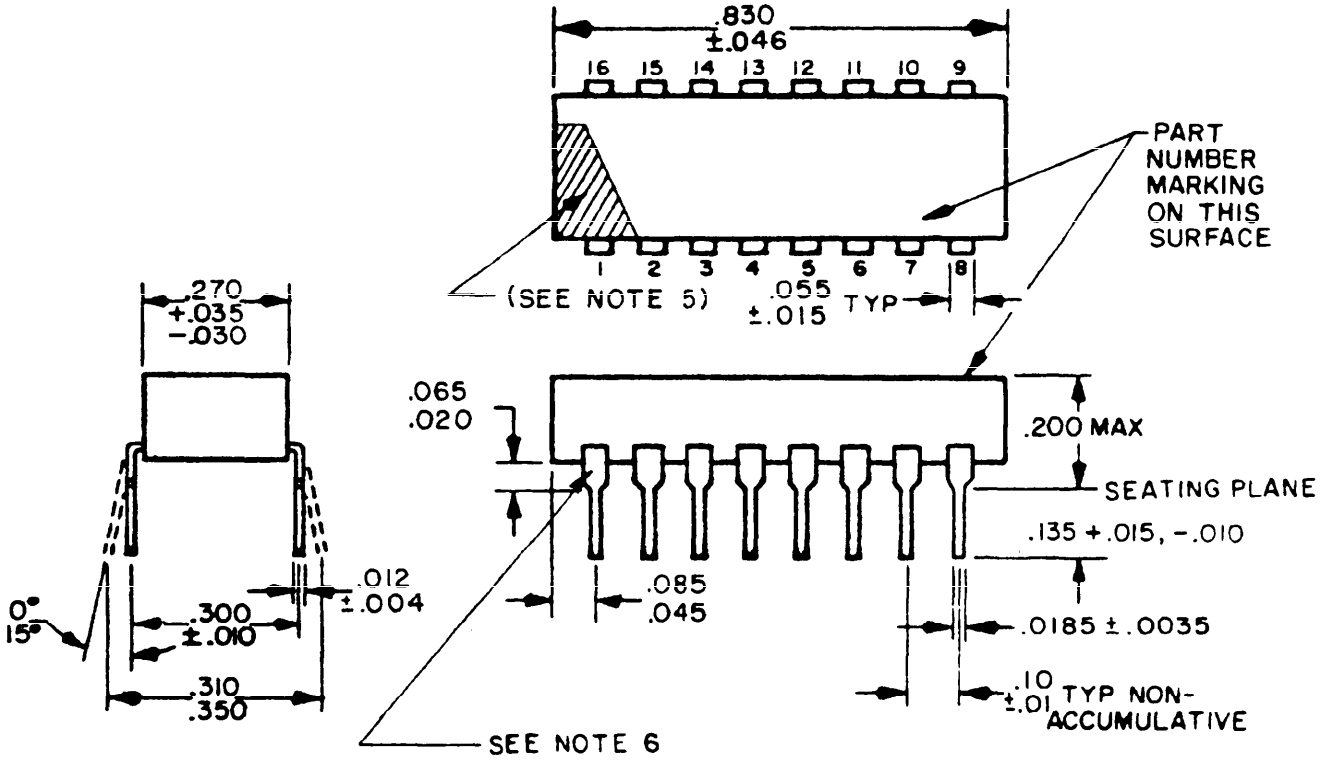


FIGURE 1. Schematics.

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Inches	mm	Inches	mm
.0035	0.089	.046	1.17
.004	0.10	.055	1.40
.010	0.25	.065	1.65
.012	0.30	.085	2.16
.015	0.38	.100	2.54
.0185	0.407	.135	3.43
.020	0.51	.200	5.08
.0225	0.572	.270	6.86
.030	0.76	.300	7.62
.035	0.89	.310	7.87
.0425	1.080	.350	8.89
.045	1.14	.830	21.08

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. The picturization of the styles above are given as representative of the envelope of the item. SLight deviations from the outline shown, which are contained within the envelope, and do not alter the functional aspects of the device are acceptable.
5. Pin 1 locator shall be a dot, notch, or numeral 1 adjacent to pin No. 1, in the shaded area.
5. Standoff shall be solderable to .010 inch above seating plane.

FIGURE 2. Style RZ110 package configuration.

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3.4.5 Dielectric withstanding voltage. When tested as specified in 4.2.3, there shall be no evidence of mechanical damage, arcing, or breakdown.

3.4.6 Insulation resistance. When tested as specified in 4.2.4, the insulation resistance of the capacitor shall be not less than 100,000 megohms unless otherwise specified.

3.4.7 Capacitance-temperature characteristic. When tested as specified in 4.2.5, the capacitance change shall not exceed ±15 percent at -55°C and +125°C when tested with minimum voltage and +15 percent, -25 percent at -55°C and +125°C when tested at rated voltage.

3.5 Network requirements. When tested as specified in MIL-R-83401 and herein, and unless otherwise specified, the networks shall meet the respective requirements of MIL-R-83401 and the following:

ΔC shall not exceed ±10 percent.

DF shall not exceed 2.5 percent.

3.5.1 Thermal shock. When tested as specified in 4.2.6, networks shall meet the requirements of 3.5.

3.5.2 Power conditioning. When tested as specified in 4.2.7, networks shall meet the requirements of 3.5.

3.5.3 Low temperature operation. When tested as specified in 4.2.8, networks shall meet the requirements of 3.5.

3.5.4 Short-time overload. When tested as specified in 4.2.9, networks shall meet the requirements of 3.5.

3.5.5 Terminal strength. When tested as specified in 4.2.10, networks shall meet the requirements of 3.5.

3.5.6 Resistance to soldering heat. When tested as specified in 4.2.11, networks shall meet the requirements of 3.5.

3.5.7 Moisture resistance. When tested as specified in 4.2.12, networks shall meet the requirements of 3.5 and the following:

Cap. DWV shall meet the requirements of 3.4.5.

Cap. IR shall be not less than 10,000 megohms.

3.5.8 Shock (specified pulse). When tested as specified in 4.2.13, networks shall meet the requirements of 3.5.

3.5.9 Vibration, high frequency. When tested as specified in 4.2.14, networks shall meet the requirements of 3.5.

3.5.10 Life. When tested as specified in 4.2.15, networks shall meet the requirements of 3.5 and the following:

Cap. DWV shall meet the requirements of 3.4.5.

Cap. IR shall be not less than 10,000 megohms.

3.5.11 High temperature exposure. When tested as specified in 4.2.16, networks shall meet the requirements of 3.5.

3.5.12 Low temperature storage. When tested as specified in 4.2.17, networks shall meet the requirements of 3.5.

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3.5.13 Solderability. When tested as specified in MIL-R-83401, networks shall meet the requirements of MIL-R-83401.

3.5.14 Resistance to solvents. When tested as specified in MIL-R-83401, networks shall meet the requirements of MIL-R-83401.

3.5.15 Network dielectric withstanding voltage. When tested as specified in MIL-R-83401, networks shall meet the requirements of MIL-R-83401.

3.5.16 Network insulation resistance. When tested as specified in MIL-R-83401, networks shall meet the requirements of MIL-R-83401.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-83401, except as noted herein.

4.1.1 Qualification inspection. Qualification inspection requirements shall consist of the following:

a. Qualification to schematic "B" of MIL-R-83401/2.

b. The inspections and number of samples specified in table II in the order shown.

4.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table III, and shall be made on the same set of sample units, in the order shown.

4.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table IV, in the order shown. They shall be performed on sample units that have been subjected to and have passed group A inspection.

4.1.4 Group C inspection. Group C inspection shall consist of the tests specified in table V, in the order shown. They shall be performed on sample units selected from lots that have passed groups A and B inspection.

TABLE II. Qualification inspection.

Inspection	Number of sample units	Requirement paragraph	Method paragraph	Number of failures allowed
Group I				
Thermal shock	All sample units	3.5.1	4.2.6	2/
Power conditioning		3.5.2	4.2.7	
DC resistance		3.3.1	4.6.5 1/	
Capacitance		3.4.1	4.2.1	
Dissipation factor		3.4.3	4.2.2	

See footnotes at end of table.

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TABLE II. Qualification inspection - Continued.

Inspection	Number of sample units	Requirement paragraph	Method paragraph	Number of failures allowed
Group II				
Resistance temperature-characteristics	10	3.3.3	4.6.8 1/	1
Capacitance temperature-characteristics		3.4.7	4.2.5	
Low temperature operation		3.5.3	4.2.8	
Short time overload		3.5.4	4.2.9	
Terminal strength		3.5.5	4.2.10	
Group III				
Thermal shock	10	3.5.1	4.2.6	
Dielectric withstanding voltage (network)		3.5.15	4.6.12 1/	
Dielectric withstanding voltage (capacitor)		3.4.5	4.2.3	
Insulation resistance (network)		3.5.16	4.6.13 1/	
Insulation resistance (cap)		3.4.6	4.2.4	
Resistance to solder heat 3/		3.5.6	4.2.11	
Moisture resistance		3.5.7	4.2.12	
Group IV				
Snock (specified pulse)	10	3.5.8	4.2.13	1
Vibration, high frequency		3.5.9	4.2.14	
Group V				
High temperature exposure	10	3.5.11	4.2.16	
Low tempearature storage		3.5.12	4.2.17	

1/ See MIL-R-83401.
2/ Networks shall meet all requirements of group I before subjecting to groups II through V.
3/ The internal inspection shall be performed after the final inspection of the group.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure	
Subgroup 1				
Thermal shock	3.5.1	4.2.6	100-percent inspection	
Power conditioning	3.5.2	4.2.7		
DC resistance	3.3.1	4.6.5 1/		
Capacitance	3.4.1	4.2.1		
Dissipation factor	3.4.3	4.2.2		
			AQL (percent defective)	
Subgroup 2			Major	Minor
Visual and mechanical inspection	3.1,3.3, 3.4,3.27, 3.28 1/	4.6.2 1/	1	4

1/ See MIL-R-83401.

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TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)	Number of failures allowed
Subgroup 1				
Resistance temp. characteristic	3.3.3	4.6.8 1/	2.5	---
Capacitance temp. characteristic	3.4.7	4.2.5		
Dielectric withstanding voltage (network)	3.5.15	4.6.12 1/		
Dielectric withstanding voltage (capacitor)	3.4.5	4.2.3		
Insulation resistance (network)	3.5.16	4.6.13 1/		
Insulation resistance (capacitor)	3.4.6	4.2.4		
Short-time overload	3.5.4	4.2.9		
Subgroup 2				
Solderability	3.5.13	4.6.6 1/	12 units	0
Resistance to solvents	3.5.14	4.6.7 1/		

1/ See MIL-R-83401.

TABLE V. Group C inspection.

Inspection	Number of sample units	Requirement paragraph	Test method paragraph	Number of failures allowed
Monthly				
Low temperature operation	10	3.5.3	4.2.8	1
Short-time overload		3.5.4	4.2.9	
Terminal strength		3.5.5	4.2.10	
Quarterly				
Subgroup 1				
Thermal shock	10	3.5.1	4.2.6	1
Dielectric withstanding voltage (network)		3.5.15	4.6.12 1/	
Dielectric withstanding voltage (capacitor)		3.4.5	4.2.3	
Insulation resistance (network)		3.5.16	4.6.13 1/	
Insulation resistance (capacitor)		3.4.6	4.2.4	
Resistance to soldering heat 2/		3.5.6	4.2.11	
Moisture resistance		3.5.7	4.2.12	
Subgroup 2				
Life	10	3.5.10	4.2.15	
Semiannually				
Subgroup 1				
Shock (specified pulse)	10	3.5.8	4.2.13	1
Vibration, high frequency		3.5.9	4.2.14	
Subgroup 2				
High temperature exposure	10	3.5.11	4.2.16	
Low temperature storage		3.5.12	4.2.17	

1/ See MIL-R-83401.
2/ The internal inspection shall be performed after the final inspection of the subgroup.

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4.2 Methods of inspection. The methods of inspection shall be in accordance with MIL-R-83401 and as specified herein.

4.2.1 Capacitance (see 3.4.1). Capacitors shall be tested in accordance with method 305 of MIL-STD-202. The following details and exception shall apply:

a. Test frequency: 1 kHz ±100 Hz.

b. Test voltage: 1.0 ±.2 volts rms.

4.2.2 Dissipation factor (see 3.4.3). The dissipation factor shall be measured with a capacitance bridge or other suitable method at a frequency of 1 kHz ±100 Hz and a voltage of 1.0 ±.2 volts rms. The inherent accuracy of the measurement shall be ±2 percent of the reading +0.1 percent dissipation factor (absolute).

4.2.3 Dielectric withstanding voltage (capacitor) (see 3.4.5). The capacitor shall be tested in accordance with method 301 of MIL-STD-202. A voltage of 2.5 times rated voltage shall be applied across the capacitor from 0 to maximum within 1 second with the surge current limited to 50 mA. The voltage shall be applied 5 ±1 seconds.

4.2.4 Insulation resistance of the capacitor (see 3.4.6). Capacitors shall be tested in accordance with method 302 of MIL-STD-202. Test voltage shall be 25 volts.

4.2.5 Capacitance-temperature characteristic (see 3.4.7). The temperature of each capacitor shall be varied as specified in table VI. Capacitance measurements shall be made at the frequency and voltage specified in 4.2.1a and b. The dc rated voltage specified in table VI shall be maintained on the capacitor during steps E to G inclusive. Capacitance measurements shall be made at each step specified in table VI and at a sufficient number of intermediate temperatures, between steps B and G to establish a true characteristic curve. Capacitors shall be kept at each temperature until temperature equilibrium is attained.

TABLE VI. Voltage-temperature limit cycle.

Step	Voltage, dc	Temperature
		°C
A	None	+25 ±2
B	None	-55 ±2
C 1/	None	+25 ±2
D	None	+125 ±2
E	Rated	+125 ±2
F	Rated	+25 ±2
G	Rated	-55 ±2

1/ Reference point.

4.2.6 Thermal shock (see 3.5.1). Networks shall be tested in accordance with MIL-R-83401 except that following dc resistance measurements, capacitance and dissipation factor shall be measured as specified in 4.2.1 and 4.2.2.

4.2.7 Power conditioning (see 3.5.2). Networks shall be tested in accordance with MIL-R-83401. The following details and exceptions shall apply:

a. Initial measurements: In addition to dc resistance, capacitance and dissipation factor shall be measured as specified in 4.2.1 and 4.2.2.

b. Operating conditions: In addition to the conditions specified, the voltage applied to the capacitor shall be 50 volts dc.

4.2.8 Low temperature operation (see 3.5.3). Networks shall be tested in accordance with MIL-R-83401 except that following dc resistance, capacitance and dissipation factor shall be measured as specified in 4.2.1 and 4.2.2.

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4.2.9 Short-time overload (see 3.5.4). Networks shall be tested in accordance with MIL-R-83401 except that following dc resistance, capacitance and dissipation factor shall be measured as specified in 4.2.1 and 4.2.2.

4.2.10 Terminal strength (see 3.5.5). Networks shall be tested in accordance with MIL-R-83401 except that following dc resistance, capacitance and dissipation factor shall be measured as specified in 4.2.1 and 4.2.2.

4.2.11 Resistance to soldering heat (see 3.5.6). Networks shall be tested in accordance with MIL-R-83401 except that following dc resistance, capacitance and dissipation factor shall be measured as specified in 4.2.1 and 4.2.2.

4.2.12 Moisture resistance (see 3.5.7). Networks shall be tested in accordance with MIL-R-83401. The following details shall apply:

a. Loading voltage: In addition to the power applied to the resistor elements, rated voltage shall be applied to the capacitor during the specified loading voltage steps.

b. Measurements: Following dc resistance measurements, capacitance and dissipation factor shall be measured as specified in 4.2.1 and 4.2.2. Following DWV of the network, capacitance DWV shall be measured as specified in 4.2.3. Following network IR measurement, capacitor insulation resistance shall be measured as specified in 4.2.4.

4.2.13 Shock (specified pulse) (see 3.5.8). Networks shall be tested in accordance with MIL-R-83401 except that following dc resistance measurement, capacitance and dissipation factor shall be measured as specified in 4.2.1 and 4.2.2.

4.2.14 Vibration, high frequency (see 3.5.9). Networks shall be tested in accordance with MIL-R-83401 except that following dc resistance measurement, capacitance and dissipation factor shall be measured as specified in 4.2.1 and 4.2.2.

4.2.15 Life (see 3.5.10). Networks shall be tested in accordance with MIL-R-83401. The following additional details shall apply:

a. Operating conditions: In addition to the power applied to each resistor element, 2X rated voltage shall be applied to the capacitor during the voltage "on" time.

b. Measurements: Following dc resistance measurements, capacitance and dissipation factor shall be measured as specified in 4.2.1 and 4.2.2. Upon completion of the life test, capacitor DWV and capacitor IR shall be measured as specified in 4.2.3 and 4.2.4.

4.2.16 High temperature exposure (see 3.5.11). Networks shall be tested in accordance with MIL-R-83401 except that following dc resistance measurements, capacitance and dissipation factor shall be measured as specified in 4.2.1 and 4.2.2.

4.2.17 Low temperature storage (see 3.5.12). Networks shall be tested in accordance with MIL-R-83401 except that following dc resistance measurements, capacitance and dissipation factor shall be measured as specified in 4.2.1 and 4.2.2.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-83401.

6. NOTES

6.1 Notes. The notes specified in MIL-R-83401 are applicable to this specification.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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